

VECTO 4.x Release Notes



VECTO v4.3.3 Official Release (04-03-2025)

Bug Fixes

- Check if XML element is signed (#950) (vecto/vecto!314)
- NgTankSystem optional for HEV lorries MRF XSD (vecto/vecto!315)
- fix: monitoring report for dual fuel vehicles (vecto/vecto!318)
- fix: secure XML loading against external entity injection (vecto/vecto!319)
- fix: correct interim supercap reader type (vecto/vecto!320)
- fix: parameter IDs for XSDs v2.3 and v2.6 (vecto/vecto!322)

Documentation

- Update XSD parameter IDs documentation (vecto/vecto!327)

VECTO v4.3.3 Official Release (03-03-2025)

Bug Fixes

- Check if XML element is signed (#950) (vecto/vecto!314)
- NgTankSystem optional for HEV lorries MRF XSD (vecto/vecto!315)
- fix: monitoring report for dual fuel vehicles (vecto/vecto!318)
- fix: secure XML loading against external entity injection (vecto/vecto!319)
- fix: correct interim supercap reader type (vecto/vecto!320)
- fix: parameter IDs for XSDs v2.3 and v2.6 (vecto/vecto!322)

VECTO v4.3.2-RC (06-02-2025)

Features

- New battery and supercap readers (vecto/vecto!309)

Bug Fixes

- ATShiftStrategyOptimized - No UpshiftFomL if not locked (vecto/vecto!301)
- Forbid downshift to locked gear in APT-S if it generates direct upshift condition (vecto/vecto!307)
- Add condition to write BusAuxiliaries output data in vsum (vecto/vecto!306)
- Allow old XMLs for battery and supercap in development only (vecto/vecto!311)
- Correct binding for supercap input data class; use correct xml data type in xml component reader for supercap (vecto/vecto!310)

VECTO v4.2.7 Official Release (09-01-2025)

Bug Fixes

- Track release_notes.md for release
- CodeEU #858: Convert steering pump tech (vecto/vecto!303)
- Converter Tool: ngTankSystem for dual fuel (vecto/vecto!304)
- CodeEU #836: Restrictions on IEPC gear and MaxTorqueCurve XSD attributes (vecto/vecto!302)

VECTO v4.2.6-RC (06-12-2024)

Features

- Support Gitlab issue pattern (vecto/vecto!272)
- CodeEU #854: Verify primary bus VIF hash against Job (vecto/vecto!295)
- CodeEU #833, #834, #835, #836, #837: Add 3rd amendment XSD definitions (vecto/vecto!291)
- CodeEU #838: Include Engine into v2.6 (vecto/vecto!297)

Bug Fixes

- Homogenize versions across tools (vecto/vecto!270)
- CodeEU #807: Produce same data from ADC loss map (#807) (vecto/vecto!278)
- Authors and readme metadata content (vecto/vecto!277)
- CodeEU #809: Writing engine information in MRF (#809) (vecto/vecto!275)
- CodeEU #842, #840, #841, #839, #798: Driver model: in case of an APT vehicle where the driving action is Brake... (vecto/vecto!287)
- CodeEU #812, #788: Conversion of doubles for SI (vecto/vecto!283)
- CodeEU #750, #758, #769, #816, #821, #829: Avoid wrong upshift and downshift for light SMT vehicles (vecto/vecto!290)
- CodeEU #844, #705, #530: During a coasting action (look-ahead coasting) a gear hunting occurs in the... (vecto/vecto!289)

- CodeEU #784: Override DoWriteModalResult for VTP (vecto/vecto!294)
- CodeEU #495, #642, #739: Add SMT downshift condition - DroppedSpd>DisengSpd (vecto/vecto!293)

VECTO v4.2.5 Official Release (02-10-2024)

Hot Fixes

- Missing Build.props DefineConstants (!268)
- Version 4th number read from Build.props (!269)

VECTO v4.2.3 Official Release (01-10-2024)

Enhancements

- CodeEU #799: Adapt VECTO for the new CI updates (!263)

Bug Fixes

- CodeEU #794: Added missing monitoring report file (!265)
- CodeEU #780: Update weights for bus subgroups (!264)

Vecto 4.2.2.3539 RELEASE CANDIDATE (09-09-2024)

Bug Fixes

- CodeEU-710: Hashing tool check fail with VECTO version 3330
- CodeEU-711: Hashing tool check fail with VECTO 4.1.3.3415
- CodeEU-712: VECTO VTP error
- CodeEU-754: "Failed to find operating point"; "Failed to find mechanic power for given electric power" in E2 vehicle
- CodeEU-727: Failure in simulating HEV in different VECTO versions
- CodeEU-749: Double summary for electric vehicles
- CodeEU-542: IVECO confidential : BUG REPORT : CRW LE T7D VOITH NXT 5.63
- CodeEU-663: IHPC: Failed to find operating point
- CodeEU-634: Article 10(2) issue - VIN YS2G6X20002202570
- CodeEU-671: IHPC: simulation abort due to unexpected response

Vecto 4.2.1.3469 OFFICIAL RELEASE (01-07-2024)

Features

- CodeEU-726: Build an XML converter tool for older VECTO jobs

Bug Fixes

- CodeEU-719: the six new tyre dimensions from line 126 onwards to the latest "wheels.csv" file in the VECTO repository
- CodeEU-717: VECTO-4.2.0.3448-RC - Buses AMT Gearbox Type with 1% higher CO2 in primary results
- CodeEU-724: Error in Primary Bus Simulation: Object reference not set to an instance of an object
- CodeEU-716: VECTO-4.2.0.3448-RC - Buses Result Summary section missing in RLST_Customer.xml
- CodeEU-694: Primary and Completed heavybus FCV article 9 exempted hashcode mismatch.
- CodeEU-735: SMT strategy different between engineering and declaration mode
- CodeEU-736: Existing customer reports (CIF) fail validation
- CodeEU-737: Missing data from XML report

Vecto 4.2.0.3448 RELEASE CANDIDATE (10-06-2024)

Features

CodeEU-696: Double summary in CIF for vocationals and non vocational missions.

CodeEU-698: Incorporate missions RD, LH and EMS to class 16 vehicles

CodeEU-697: Re-evaluate subgroup allocations for Long Haul

CodeEU-676: Feature: Implement monitoring report

Bug fixes (1/4)

- CodeEU-471: VectoSimulationException: VF640J869RB022573
- CodeEU-462: Article10-2-issue | Order-Nr 28206354 | HEV P1 Error Mercedes-Benz and Setra Hybrid
- BusesCodeEU-326: Article10-2-issue | Order-Nr 28195581 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-348: Article10-2-issue | Order-Nr 28204519 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-457: Article10-2-issue | Order-Nr 28174000 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-458: Article10-2-issue | Order-Nr 28186528 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-459: Article10-2-issue | Order-Nr 28203057 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-545: IVECO confidential : BUG REPORT : HEV-P1 : UW18m C9 VOITH NXT CRU 48V mild hybrid
- CodeEU-346: Article10-2-issue | Order-Nr 28202338 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-352: Article10-2-issue | Order-Nr 28202130 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

- CodeEU-394: Article10-2-issue | Order-Nr 28204065 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (2/4)

- CodeEU-433: Article10-2-issue | Order-Nr 28192321 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-451: Article10-2-issue | Order-Nr 28204280 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-452: Article10-2-issue | Order-Nr 28197394 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-453: Article10-2-issue | Order-Nr 28199435 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-454: Article10-2-issue | Order-Nr 28206982 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-655: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208126 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-660: Retarder lossmap error in completed vehicle simulation
- CodeEU-662: Generic retarder map speed range insufficient in some cases
- CodeEU-648: Error in Multistep Tool PEV/P-HEV
- CodeEU-618: PEV vehicles simulation error depending the time format
- CodeEU-700: Factor Method Generic IHPC Powermap De-normaization bug
- CodeEU-482: Article10-2-issue | Order-Nr 28203040 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-514: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208051 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-529: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28199994 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (3/4)

- CodeEU-544: IVECO confidential : BUG REPORT : HEV-P1 : CRW LE C9 VOITH NXT CRU 48V mild hybrid
- CodeEU-552: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28201759 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-556: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208176 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-557: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28209751 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-622: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28210594 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-632: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28210591 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

- CodeEU-672: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28208841 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-673: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209179 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-674: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28211540 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (4/4)

- CodeEU-678: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28192673 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-697: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209551 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-685: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28200286 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-686: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28201178 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-687: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209679 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-692: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209789 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Vecto 4.1.3.3415 OFFICIAL RELEASE (08-05-2024)

Hot fixes

- CodeEU-638: Incorrect construction payloads for group 9 non-vocational vehicle

Bug fixes

- CodeEU-615: Multistep freezes after loading VIF chassis
- CodeEU-616: Multistep tool freezes
- CodeEU-619: restore wrong exempted techs in XSD for backwards compatibility
- CodeEU-617: Results change depending the time format
- CodeEU-635: Revert multiple summary in CIF

Vecto 4.1.0.3392 RELEASE CANDIDATE (15-04-2024)

Features

- CodeEU-577: Add missing mission profiles to vocational and non-vocational

Bug fixes (1/4)

- CodeEU-331: Gear 1 DrivingActionAccelerate: Fail
- CodeEU-367: Gear 1 DrivingActionAccelerate: Fail
- CodeEU-372: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-373: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-374: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-375: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-393: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-439: 615 (Interurban _P32DD_ReferenceLoad) - absTime: 7129.6359 [s], distance: 53875.9765 [m], dt: 0.6388 [s], v: 0.5098 [m/s], Gear: 1 | DrivingActionAccelerate: Failed
- CodeEU-446: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-449: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-478: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to find operating poi

Bug fixes (2/4)

- CodeEU-481: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 |
- CodeEU-488: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to find operating poi
- CodeEU-494: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 26 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to fi
- CodeEU-420: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-573: Unhandled powertrain architecture Conventional Vehicle to calculate gradability
- CodeEU-582: unhandled powertrain architecture ConventionalVehicle to calculate gradability, 11:44:36.63,
- CodeEU-595: Inconsistency exempted vehicles "Fuel cell vehicle" vs. "FCV Article 9 exempted"
- CodeEU-501: Help regarding VTP vdri format is not up to date in v4
- CodeEU-594: inadequate validation for auxiliaries in completed vehicle XML
- CodeEU-580: Potential error in XML schema v2.4 (exempted vehicles)
- CodeEU-551: Article-10-2 XLRASF5E00G419905
- CodeEU-547: Max ICE Off timespan for buses
- CodeEU-583: Incorrect internal resistance for SuperCap used in factor method

Bug fixes (3/4)

- CodeEU-571: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28202896 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-526: VTP calculation aborted on fuel consumption map
- CodeEU-502: Error in VECTO calculation WMA10CZZ0RF022326
- CodeEU-401: Electric steering system with conventional vehicle not according to 2017/2400
- CodeEU-476: Setting Initial SoC in REESS editor leads to error

- CodeEU-533: Question about heavy lorry_IEPC_Gb×4speed: Failed to generate electric power map
- at least two negative entries are required
- CodeEU-456: Clarification Documentation official Results
- CodeEU-506: Article 10(2) issue - VIN YS2P6X200R2201285
- CodeEU-507: Article 10(2) issue - VIN YS2P6X20005732399
- CodeEU-508: Article 10(2) issue - VIN S2P6X20005734199
- CodeEU-499: Object Reference not set Error
- CodeEU-546: Problem with P181 (Cooling Fan Technology) for a conventional vehicle
- CodeEU-516: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208044 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-517: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208037 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (4/4)

- CodeEU-519: Article10-2-issue | VECTO-4.0.3 | Order-Nr 10098181 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-527: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28196233 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-528: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28196652 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-338: Eco-roll only, without engine stop, impact on CO2 emission
- CodeEU-475: XML files loading in VTP Job not OK; No auxiliary data and torque curve visible, Job can't be saved in VECTO 4.0.3.3330
- CodeEU-610: VTP mode is broken due to changes in the Clutch component.

Vecto 4.0.3.3330 OFFICIAL RELEASE (13-02-2024)

• Bug Fixes

- CodeEU-293: DistanceRun got an unexpected response
- CodeEU-298: Object reference not set to an instance of an object
- CodeEU-300: Full drive torque miscalculation
- CodeEU-336: Feature: Hashing tool must validate the previous step data for multi-step jobs
- CodeEU-337: Electric Citybus - ERROR with "31a-Specific" bus configuration
- CodeEU-343: Cannot simulate Primary Vehicle using vectocmd.exe
- CodeEU-387: VECTO sometimes fails to properly read Tyre data from primary vehicle xml
- CodeEU-427: Vehicle speed resulting to exceeding max gearbox speed
- CodeEU-428: Remove speed safety margin for gearbox re-engaging
- CodeEU-438: Mismatch XML schema vs Regulation (exempted vehicles)
- CodeEU-248: Bus P2 hybrid VECTO error in urban cycle
- CodeEU-278: Simulation crash when writing fuel consumption results to reports
- CodeEU-284: New error message and failed simulation obtained for P2 hybrid buses
- CodeEU-285: PEV vehicle error in routine to write the results with new vecto version 4.0.2.3275
- CodeEU-287: PEV_IEPC error message:"can only operate on SI Objects with the same unit"
- CodeEU-289: VECTO Simulation Error for bus with validated input data

Vecto 4.0.2.3275 OFFICIAL RELEASE

(20-12-2023)

Hot fix

- CodeEU-273, CodeEU-274: Changes in the AMT shift strategy regarding idling speed caused simulation aborts
- CodeEU-260: regression fix handling overload buffer

Bug fixes (1/2)

- CodeEU-94: DistanceRun got an unexpected response
- CodeEU-153: DrivingActionAccelerate: Failed to find operating point after Overload
- CodeEU-158: Urban RefLoad DrivingActionAccelerate: Failed to find operating point (IEPC Wheelhub 1 measured)
- CodeEU-168: Vecto Declaration Simulation with P1-Hybrid shows multiple errors - Simulation Aborts!
- CodeEU-191: Boosting limits HEV ovc are not working
- CodeEU-202: EMS Standard Values: Continuous/Overload Torque 0 Nm
- CodeEU-203: HybridStrategy error for IHPC type 1 hybrid lorry
- CodeEU-206: VTP + PEMS test done by Renault Trucks France - VECTO tool errors most probably linked to automatic gearbox (with torque converter)
- CodeEU-211: Hybrid P1 configurations with errors
- CodeEU-215: MultiStep tool help
- CodeEU-216: NrOfGears in CIF is 1 for IEPC no matter how many gears it has
- CodeEU-220: Error manual transmission 2. gear
- CodeEU-224: Issue with PEV complete vehicle simulation (Vecto MultiStage)
- CodeEU-231: IVECO CONFIDENTIAL : hybrid buses completed simulation aborted
- CodeEU-234: Error Conventional Lorry Gear: 1C DistanceRun got an unexpected response
- CodeEU-235: Tyre error calculation buses "invalid xsi:type 'TyreDataDeclarationType'"
- CodeEU-238: Bus VIF files not valid in Multistep VECTO
- CodeEU-243: Simulation aborted: Gear 5 Lossmap not sufficient
- CodeEU-244: Signature validation fails for old (prior to v4) Manufacturer reports

Bug fixes (2/2)

- CodeEU-249: Fix handling gear torque limits in case of IEPC WheelHub motor and only one side is measured
- CodeEU-250: Maximum vehicle speed exceeded during pre-processing
- CodeEU-253: Replace VTP HeavyBus in Generic Vehicles with VTP Truck
- CodeEU-259: Inconsistent calculation of AverageRRC in CIF
- CodeEU-260: Handling of overload buffer in case Continuous Torque is 0Nm
- CodeEU-261: SuperCap internal resistance standard values correction
- CodeEU-263: Determination of rated power for IEPCs for MRF and CIF
- CodeEU-266: Fix Measured Speed Testcases

Vecto 4.0.1.3217 OFFICIAL RELEASE (23-10-2023)

Improvements

- VTP mode: create zip archive for input and output files

Bug Fixes

- bugfix in CIF for complete(d) buses - do not write PrimaryVehicleSubgroup element
- fix hybrid strategy: power comparison
- fix exempted vehicles do not work (error message that XML version is not supported)
- bugfix in XML schema: remove wrong PS technology entry for lorries
- bugfix for conditioning power demand for IHPC vehicles

VECTO 4.0.0.3210 Release Notes (16-10-2023)



Release Notes

First official VECTO release for the 2nd Amendment of Regulation (EU) 2017/2400

Features

- Declaration Mode
 - xEV-Heavy Lorries
 - Conventional Medium Lorries
 - xEV Medium Lorries
 - Conventional Buses
 - xEV Buses

- Dedicated user interface for simulating buses in Declaration Mode using the multistep approach: VECTOMultistep.exe
- Engineering Mode
 - xEV-Vehicles
- Updated XML-Reports and Vehicle Information Files to support new Vehicle Architectures
 - New Manufacturer Record File (MRF)
 - New Customer Information File (CIF)
 - Multistep Output (VIF):
- Updated simulation output (.vsum, .vmod)
- Updated Hashing Tool to handle new powertrain components (electric motor, battery, supercap ...)
- Supports .NET Framework 4.8 and .NET 6.0

Features - Tool modes

	Medium Lorries (XML)	Heavy Lorries (XML)	Primary buses (XML)	Interim buses (GUI)	Interim buses (XML)	Completed buses (JSON v7)	Completed buses (GUI)	Completed buses (XML)
VECTO.exe	S	S	S			S		
VECTOcmd.exe	S	S	S			S		
VECTOMultistep.exe	S	S	S	S+E	S+E	S+E	S+E	S
Comments				function: generates updated VIF				

S: Simulation only.

S+E: Simulation and editing.

Changes

- XML Job Files < v2.4 are no longer supported
- Dropped support for .NET Framework 4.5 (EOL 04/2022)

Implementation of Declaration Mode for xEV-Vehicles

- Generic auxiliary parametrisation
- Generic parametrisation of usable SOC range
- Generic parametrisation of Serial/Parallel Hybrid Strategy parameters
- Automated simulation of OVC-HEV
 - Charge depleting mode (CD): The propulsion energy is provided by the electric storage only
 - Charge sustaining mode (CS): The propulsion energy is provided by the fuel storage

Implementation of Declaration Mode for xEV-Lorries

Generic E-PTO for PEV and S-HEV in Declaration Mode

E-PTO for PEV and S-HEV in Engineering Mode

Implementation of Declaration Mode for Buses

Factor Method for conventional and xEV-Buses

Updated bus auxiliary model